

EX. NO. 06 - SLEUTH KIT DIGITAL EVIDENCE ANALYSIS

Experiment Name	Use Sleuth Kit to Analyze Digital Evidence
Aim	To analyze a forensic disk image using The Sleuth Kit and examine the partition structure, file system, files, metadata, and recover a file from the digital evidence.
Tools Required	The Sleuth Kit 4.14.0; Windows Command Prompt; Windows File Explorer; Forensic Disk Image (4Dell Latitude CPI.E01)
Evidence Used	Evidence File: 4Dell Latitude CPI.E01

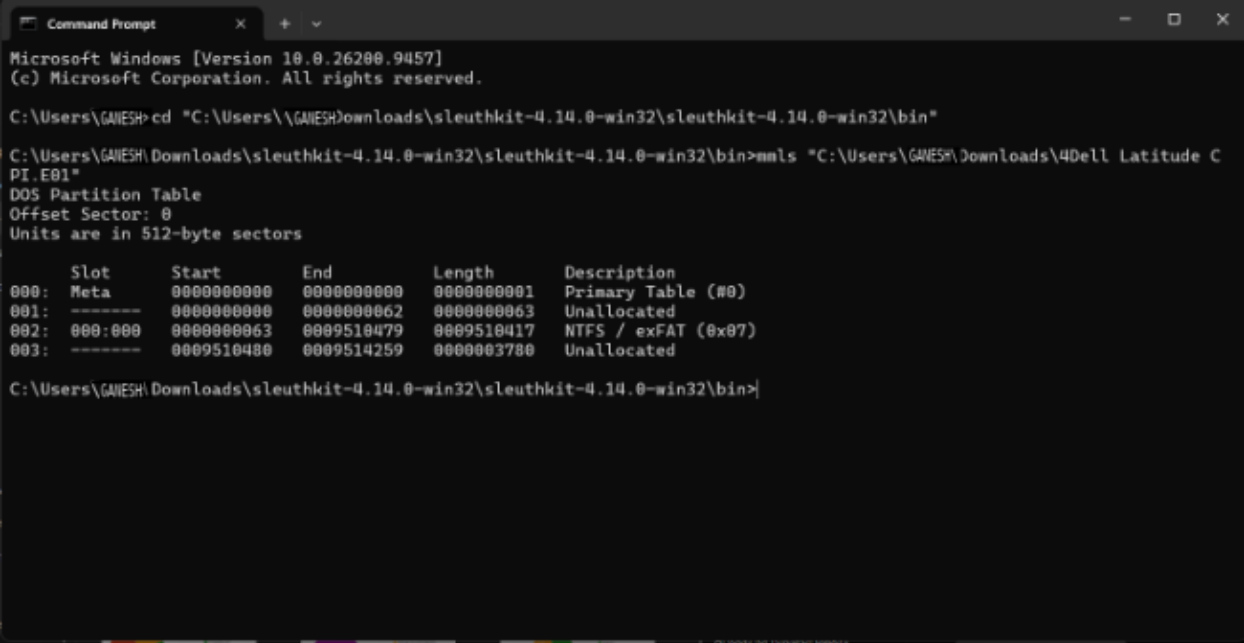
PROCEDURE AND OUTPUTS

STEP 1 — Identify Partition Structure

The mmls command was used to identify the partition structure of the forensic disk image.

Command:

```
mmls "C:\Users\GANESH\Downloads\4Dell Latitude CPI.E01"
```



```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GANESH>cd "C:\Users\GANESH\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin"

C:\Users\GANESH\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls "C:\Users\GANESH\Downloads\4Dell Latitude C
PI.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

   Slot      Start      End      Length    Description
000: Meta    0000000000  0000000000  0000000001  Primary Table (#0)
001: -----  0000000000  0000000062  0000000063  Unallocated
002: 000:000  0000000063  0009510479  0009510417  NTFS / exFAT (0x07)
003: -----  0009510480  0009514259  0000003780  Unallocated

C:\Users\GANESH\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

Figure 1: Partition structure of the forensic disk image identified using mmls

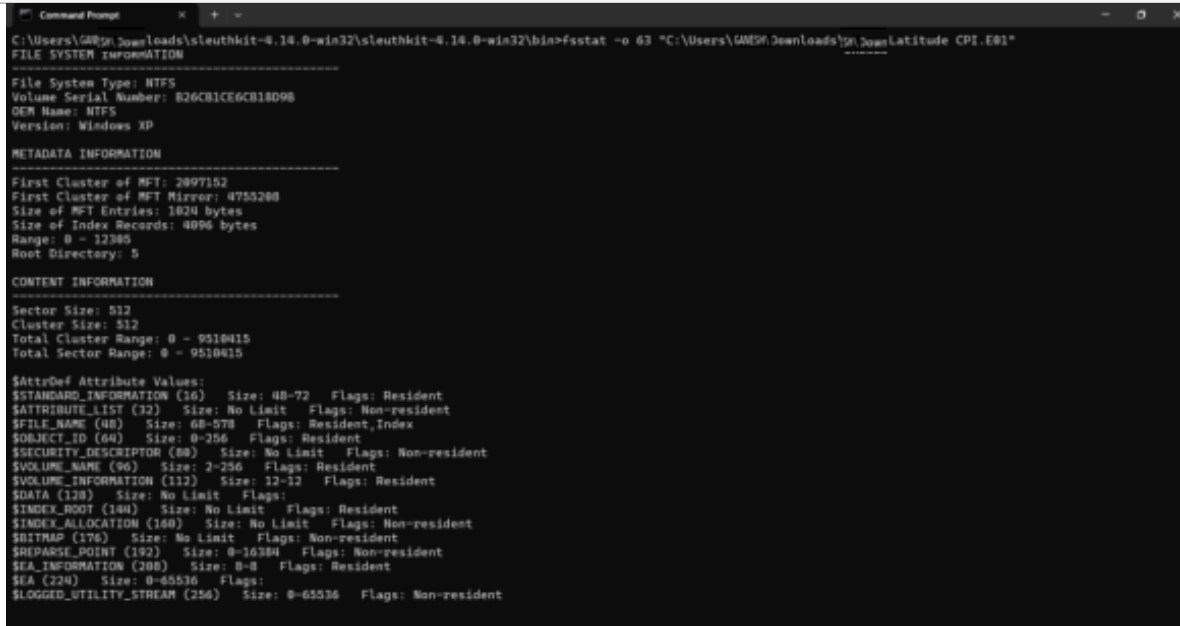
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STEP 2 — Analyze File System

The fsstat command was used to obtain detailed information about the file system.

Command:

```
fsstat -o 63 "C:\Users\GANESH\Downloads\4De11 Latitude CPI.E01"
```



```
C:\Users\GANESH\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -o 63 "C:\Users\GANESH\Downloads\4De11 Latitude CPI.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB1809B
GEN Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2897152
First Cluster of MFT Mirror: 4755248
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12385
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-576 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (88) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 0-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
```

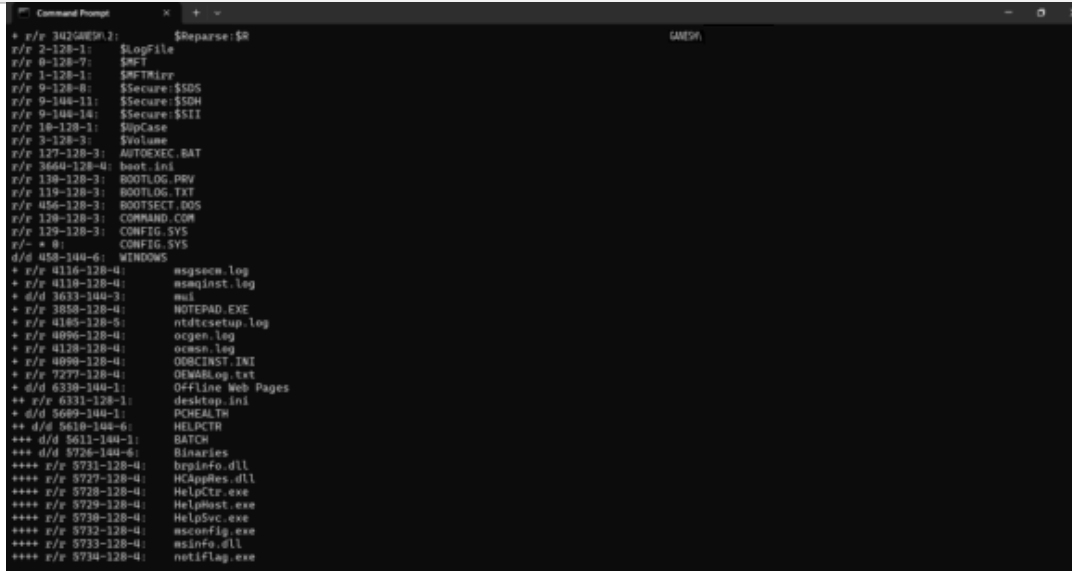
Figure 2: NTFS file system information obtained using fsstat

STEP 3 — List Files and Directories

The fls command was used to list files and directories available in the forensic disk image. The required file airplane.bmp was identified from the file listing.

Command:

```
fls -o 63 -r "C:\Users\GANESH\Downloads\4De11 Latitude CPI.E01"
```



```
+ r/r 342GANESH.2: $Reparse:$R
r/r 2-128-1: $LogFile
r/r 0-128-7: $MFT
r/r 1-128-1: $MFTMirr
r/r 9-128-8: $Secure:$DS
r/r 9-144-11: $Secure:$DSH
r/r 9-144-14: $Secure:$SII
r/r 18-128-1: $UpCase
r/r 3-128-3: $Volume
r/r 127-128-3: AUTOEXEC.BAT
r/r 3640-128-4: boot.ini
r/r 130-128-3: BOOTLOG.PRV
r/r 119-128-3: BOOTLOG.TXT
r/r 456-128-3: BOOTSECT.DOS
r/r 120-128-3: COMMAND.COM
r/r 120-128-3: CONFIG.SYS
r/r * 0: CONFIG.SYS
d/d 458-144-6: WINDOWS
+ r/r 4116-128-4: msqssec.log
+ r/r 4118-128-4: msqinst.log
+ d/d 3637-144-3: mca
+ r/r 3858-128-4: NOTEPAD.EXE
+ r/r 4185-128-5: ntdtsetup.log
+ r/r 4096-128-4: ocgen.log
+ r/r 4128-128-4: ocwsn.log
+ r/r 4099-128-4: OMCINST.INI
+ r/r 7277-128-4: OEMARLog.txt
+ d/d 6330-144-1: Offline Web Pages
++ r/r 6331-128-1: desktop.ini
+ d/d 5689-144-1: PCHEALTH
++ d/d 5618-144-6: HELPCTR
+++ d/d 5611-144-1: BATCH
+++ d/d 5726-144-6: Binaries
++++ r/r 5731-128-4: bgrinfo.dll
++++ r/r 5727-128-4: HCApplRes.dll
++++ r/r 5728-128-4: HelpCtr.exe
++++ r/r 5729-128-4: HelpInst.exe
++++ r/r 5730-128-4: HelpSec.exe
++++ r/r 5732-128-4: msconfig.exe
++++ r/r 5733-128-4: msinfo.dll
++++ r/r 5734-128-4: notiflag.exe
```

Figure 3: Files and directories listed using the fls command

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STEP 4 — Analyze File Metadata

The istat command was used to examine the metadata of the identified file through its Master File Table (MFT) entry.

Command:

```
istat -o 63 "C:\Users\GANESH\Downloads\4Dell Latitude CPI.E01" 5665
```

STEP 5 — Recover the Identified File

The icat command was used to recover the identified file from the forensic disk image using its MFT entry number.

Command:

```
icat -o 63 "C:\Users\GANESH\Downloads\4Dell Latitude CPI.E01" 5665 >
"C:\Users\GANESH\Downloads\DF_Output\airplane.bmp"
```

```
C:\Users\GANESH\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mk
dir "C:\Users\GANESH\Downloads\DF_Output"
A subdirectory or file C:\Users\GANESH\Downloads\DF_Output already exists.
```

Figure 4: Metadata and recovery command output

STEP 6 — Verify the Recovered File

The recovered airplane.bmp file was located in the DF_Output folder and opened using Windows File Explorer. The image was displayed successfully, confirming that the file was recovered from the forensic disk image.

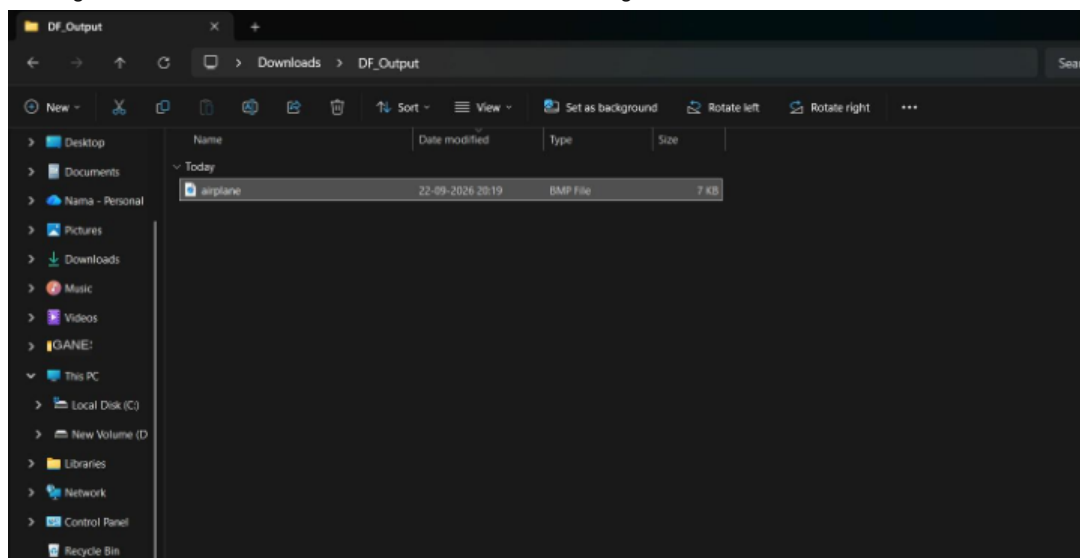


Figure 5: Recovered airplane.bmp file in the DF_Output folder

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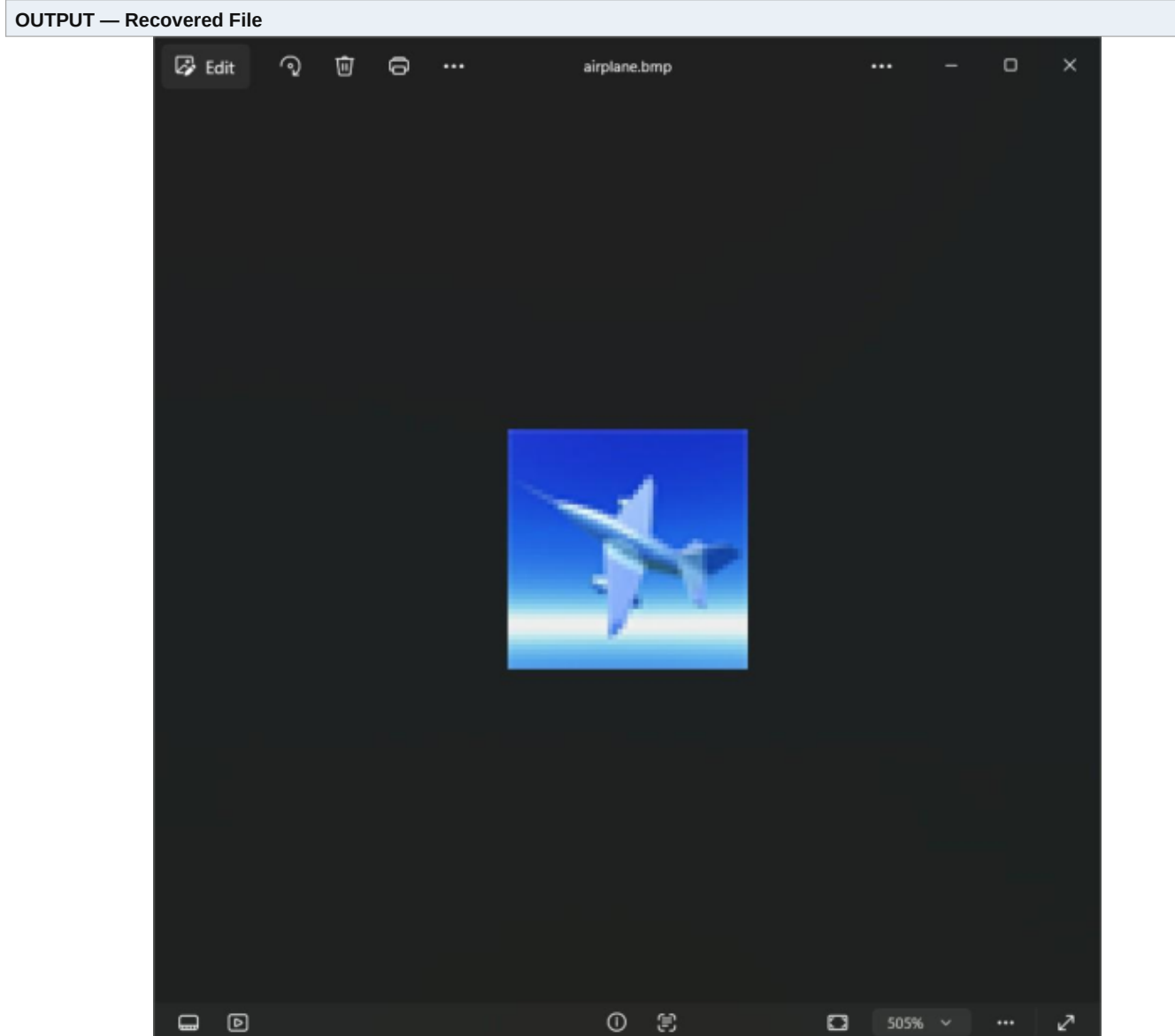


Figure 6: Recovered airplane.bmp opened successfully

RESULT

The forensic disk image was successfully analyzed using The Sleuth Kit. The partition structure and NTFS file system were examined, files and directories were listed, file metadata was analyzed, and airplane.bmp was successfully recovered.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	